

Right-Triangle Trigonometry — Wrong Ratio, Wrong Inverse, Wrong Mode

Answer Key

High School Geometry

Quick Answers

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| 1. 6.88 | 3. 16.26 | 5. 31.25 |
| 2. 17.92 | 4. 15.09 | 6. 5 |
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Curriculum

Level: High School Geometry

HSG-SRT.C.8 – problems 1, 2, 3, 4, 5, 6

Difficulty 2–5 of 5 across 6 tagged checks.

Common wrong answers

1. If they got 9.83: used cosine, which pairs the ADJACENT leg with the hypotenuse, where the opposite leg was wanted
 2. If they got -84.75: calculator left in radian mode, so it evaluated the tangent of 52 radians
 3. If they got 73.74: put the sides in upside down, taking the inverse tangent of adjacent over opposite
 4. If they got 13.12: used sine where the leg ADJACENT to the given angle was wanted
 5. If they got 12.41: calculator left in radian mode, so it evaluated the tangent of 38 radians
 6. If they got 8.66: used the sine of the other acute angle, which returns the longer leg
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1. Hypotenuse 12, $\angle A = 35^\circ$; find x opposite $\angle A$.

Step 1. Name the two sides in play. The unknown is OPPOSITE $\angle A$; the given side is the HYPOTENUSE. Opposite with hypotenuse is sine — the S of SOH-CAH-TOA.

Step 2. Set up and solve: $\sin 35^\circ = \frac{x}{12}$, so $x = 12 \sin 35^\circ$.

Step 3. In degree mode, $12 \sin 35^\circ = 6.8829\dots$, which rounds to 6.88

Step 4. Sanity check: 6.88 is shorter than the hypotenuse 12, as every leg must be. Cosine would have given 9.83 — also shorter than 12, which is exactly why the size check alone does not catch a wrong ratio. Naming the two sides first does.

2. Dana: $x = 14 \tan 52^\circ$, calculator shows -84.75 .

Step 1. Confirm the setup. The unknown is opposite $\angle A$, the given leg 14 is adjacent to it, and opposite over adjacent is tangent: $x = 14 \tan 52^\circ$ is correct.

Step 2. Diagnose the number. A length can never be negative, so the fault is in how the calculator read 52. In radian mode it evaluates the tangent of 52 radians, which is a completely different angle and can land anywhere — here at -84.75 .

Step 3. Switch the calculator to DEG (look for the DEG/RAD indicator) and recompute: $14 \tan 52^\circ = 14(1.2799\dots) = 17.9192\dots$, so $x = \boxed{17.92}$

Step 4. Sanity check: $\angle A = 52^\circ$ is bigger than 45° , so the opposite leg should be LONGER than the adjacent leg 14 — and 17.92 is.

3. Eli: $\angle A = \tan^{-1}(24/7) = 73.74^\circ$, from legs 7 opposite and 24 adjacent.

Step 1. Write the ratio from the definition before touching the inverse: $\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{7}{24}$. Eli inverted the fraction, which is the most common inverse-trig error — the calculator cannot see that the sides went in upside down.

Step 2. Apply the inverse to the correct ratio: $\angle A = \tan^{-1}\left(\frac{7}{24}\right) = 16.2602\dots$

Step 3. Rounded: $\boxed{A \approx 16.26^\circ}$

Step 4. Sanity check: the side opposite $\angle A$ is much shorter than the adjacent side, so $\angle A$ must be the SMALL acute angle — well under 45° . Eli's 73.74° is the other acute angle of this same triangle, and the two sum to 90° , which is a quick way to spot the swap.

4. Hypotenuse 20, $\angle A = 41^\circ$; find the adjacent leg b .

Step 1. Name the sides. The unknown is ADJACENT to $\angle A$ and the given side is the HYPOTENUSE, so the ratio is cosine, not sine.

Step 2. $\cos 41^\circ = \frac{b}{20}$, so $b = 20 \cos 41^\circ$.

Step 3. In degree mode: $20 \cos 41^\circ = 15.0942\dots$, so $b = \boxed{15.09}$

Step 4. The classmate's sine gives $20 \sin 41^\circ = 13.12$, which is the OTHER leg of this triangle. Both numbers are plausible lengths; only the opposite/adjacent naming separates them.

5. Flagpole: 40 ft away, angle of elevation 38° , screen shows 12.41.

Step 1. Draw the right triangle: the horizontal distance 40 is adjacent to the 38° angle, and the height is opposite it. Opposite over adjacent is tangent, so height $= 40 \tan 38^\circ$. Ravi's setup is correct.

Step 2. Test the displayed number. At 38° — close to 45° — the height should be nearly as large as the 40-foot distance. A height of 12.41 would need an angle of about 17° , so the number is wrong even though the formula is right.

Step 3. The calculator is in radian mode: it returned $40 \tan(38 \text{ rad})$. Switch to degrees and recompute: $40 \tan 38^\circ = 40(0.78128\dots) = 31.2514\dots$

Step 4. The pole rises $\boxed{31.25 \text{ ft}}$ above Ravi's eye level.

Step 5. Sanity check: $31.25 < 40$, which is right for an angle under 45° , and the ratio $31.25/40 = 0.78$ is indeed $\tan 38^\circ$.

6. 30-60-90 triangle, hypotenuse 10: find a opposite 30° , and identify Sasha's 8.66.

(a) Step 1. In a 30-60-90 triangle the sides are in the ratio $1 : \sqrt{3} : 2$, with the shortest side opposite the 30° angle and the hypotenuse opposite the right angle. So the side opposite 30° is exactly half the hypotenuse.

Step 2. $a = \frac{10}{2} = 5$, and the trigonometric route agrees: $a = 10 \sin 30^\circ = 10(0.5) = 5$,

because $\sin 30^\circ$ is exactly $\frac{1}{2}$. $a = 5$

(b) Sasha computed $10 \sin 60^\circ = 8.66$. That is the side opposite the 60° angle — the LONGER leg, $5\sqrt{3} = 8.6602\dots$ — so the number itself is a real length in this triangle, just not the one asked for. The error is the classic one: the ratio was taken from the other acute angle. Naming the angle you are working from, before choosing sin or cos, is what prevents it.

Grading note: accept "the longer leg, opposite the 60° angle, found with $\sin 60^\circ$ (or $\cos 30^\circ$)" as a full-credit answer to part (b).